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# Synthesis of superhydrophilic Gradient-Like Copolymers: Kinetics of the RAFT copolymerization of methacryloyloxyethyl phosphorylcholine with PEO methacrylate

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ABSTRACT

Brush copolymers containing PEO side chains and zwitterionic units of 2-methacryloyloxyethyl phosphorylcholine (MPC) were synthesized by RAFT copolymerization with an especial attention to compositional microstructure of the copolymers. Two complementary methods – size exclusion chromatography (*SEC*) and $^1$H NMR spectroscopy – were used to study the kinetics of the RAFT copolymerization of MPC ($M_1$) and the macromonomer poly(ethylene oxide) methyl ether methacrylate with medium-length PEO chain (PEOMEMA, $M_2$) in methanol–water mixed solutions. Analysis of the reaction mixtures by the two methods allowed concluding that in mixed solutions of methanol and water, every molecule of MPC carries about 15 molecules of water as an integral part of the monomer. Monomer reactivity ratios calculated by the method of Meyer and Lowry were $r_1 = 2.48$ and $r_2 = 0.40$. The dependences of the copolymer composition on monomer conversion determined experimentally and calculated by the terminal model of free-radical copolymerization differed significantly, demonstrating the specific impact of the mechanism of RAFT on the distribution of copolymer compositions. The copolymers p(MPC-*co*-PEOMEMA) possessed strongly expressed gradient microstructure, and were characterized by moderately low dispersity ($Đ < 1.3$). The copolymers showed evident superhydrophilicity with a water contact angle of about 7° and, like pMPC, exhibited cononsolvency in water–ethanol mixed solutions.

## 1. Introduction

Hydrophilic polymers with phosphorylcholine head groups, such as poly(2-methacryloyloxyethyl phosphorylcholine) (pMPC), are antifouling materials that are suitable for direct penetration across cell membrane due to imitation of bio-membrane structure, have the specific binding with C-reactive protein, exhibit fluid lubrication, etc. [1]. The phosphorylcholine unit present in MPC polymers makes these polymers congenerous with lipids that could be found on the surface of living cells. The phosphorylcholine polymers show excellent hydrophilicity, biocompatibility and long-term stability, and have the potential to be applied as mimetic cell membranes for bio-interfaces [2,3]. Polymers with MPC units are promising to be employed as biosensors [4], drug delivery agents, and bioinert materials [5–7]. Polymer brushes containing MPC segments could act as efficient lubricants between surfaces in aqueous media regardless of their moderate graft density [8,9]. MPC was used in surface-initiated copolymerization for implantable medical device surface modification providing advantageous hydrophilicity and antifouling properties [10,11].

Biocompatible statistical copolymers containing MPC units were typically prepared via conventional free-radical copolymerization [11–14]. However, due to the poor solubility of a comonomer and, in many cases, target copolymers in water, the copolymerization of MPC is usually conducted in organic solvents or mixtures of water and water-miscible organic solvents [14,15]. MPC copolymers with styrene [13], alkyl methacrylate [11,16] and other comonomers [17,18] were successfully synthesized and used as drug delivery systems, lubricants, and antifouling materials. Subject to comonomer, MPC was either a more active or less active monomer in free-radical copolymerization. Thus, reactivity ratios of MPC ($M_1$) in free-radical copolymerization with styrene and 2-vinyl-4,4-dimethylazlactone are $r_1 = 0.39$, $r_2 = 0.46$ [13] and $r_1 = 0.74$, $r_2 = 1.55$ [14], respectively. Much higher reactivity ratio of MPC ($M_1$) was in free-radical copolymerization with 2-methoxyethyl acrylate and 2-deoxy-2-methacrylamido-D-glucose, $r_1 = 2.21$, $r_2 = 0.53$ [11] and $r_1 = 1.43 \pm 0.27$, $r_2 = 0.40 \pm 0.14$ [12], respectively.

Some MPC containing copolymers were synthesized by methods of

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reversible deactivation (living) radical polymerization, e.g., RAFT. RAFT copolymerization of MPC and several methacrylates (2-gluconamidoethyl methacrylamide, 2-lactobionamidoethyl methacrylamide, etc.) carried out in water/propanol (87:13 v/v) mixtures, yielded 80 % monomer conversion and resulted in copolymers with relatively low dispersity index (Đ 1.2–1.25) [19]. RAFT copolymerization of MPC and 4-(N-(S-penicillaminylacetyl)amino)phenylarsonous acid methacrylamide was performed in water/methanol (1:1 v/v) solutions resulting in the copolymers with low molecular weight ($M_n < 13$ kDa) and low dispersity index (Đ 1.12–1.14) [20]. RAFT copolymerization of MPC and 3-formyl-4-hydroxybenzyl methacrylate was carried out in anhydrous DMF, yielding fluorescent polymeric nanoparticles [21]. Despite recent reports on the synthesis of MPC-containing macromolecules, development of novel polymeric structures (e.g., brush-type) and looking forward to their potential application in bio-related fields remains in high demand.

To the best of our knowledge, there are no published data about the copolymerization of MPC and poly(ethylene glycol) methyl ether methacrylates (PEOMEMA), neither by conventional free-radical copolymerization nor by RAFT copolymerization. Moreover, there are only a few data about the reactivity ratio of MPC in free-radical copolymerization, and no data about reactivity ratios of MPC and comonomers in RAFT copolymerization.

The purpose of the present study was to demonstrate the possibilities of synthesizing well-defined brush copolymers containing medium-length PEO side chains and zwitterionic MPC units by RAFT copolymerization. An especial focus was directed to the kinetics of the RAFT copolymerization of MPC and PEOMEMA, which is very important in elucidating if the copolymers are statistical or gradient. Moreover, the present study demonstrates that size exclusion chromatography used simultaneously with $^1$H NMR spectroscopy provides valuable information about changes in macromolecular composition during RAFT copolymerization.

## 2. Materials and methods

### 2.1. Materials

Poly(ethylene oxide) monomethyl ether methacrylate (PEOMEMA, $M_n$ 950, Aldrich) was purified from inhibitors by dissolving in water (1:1) and passing through a chromatographic column filled with basic alumina (Type 5016A, Fluka). Methanol (>99.9 %, Merk) and other solvents (ethyl acetate, hexane, diethyl ether (Chempur)) were used without further purification. 4,4-Azobis(4-cyanovaleric acid) (ACVA, 98 %, Fluka), 2-methacryloyloxyethyl phosphorylcholine (98 %, Merk) were used as received. RAFT chain transfer agent 4-(((butylthio)carbonothioyl)thio)-4-cyanopentanoic acid (CTA) was synthesized just before polymerization according to a previously published procedure with minor adjustments [22].

### 2.2. RAFT copolymerization of MPC and PEOMEMA

For the synthesis of the copolymers, MPC and PEOMEMA were copolymerized at different molar ratios (75:25, 50:50, and 25:75, respectively). The initial molar ratio of CTA to the initiator was kept constant ($[CTA]_0/[I]_0 = 3/1$). The copolymerization procedure of MPC and PEOMEMA at an initial monomer molar ratio of 50:50 is presented below ($[M]_0/[CTA]_0/[I]_0 = 240/3/1$): MPC (0.59 g, 2.0 mmol), PEOMEMA solution (3.8 g, 2.0 mmol), RAFT CTA (14.5 mg, 0.05 mmol) and ACVA (4.67 mg, 0.0166 mmol) were placed in 50 mL round-bottom flask and dissolved in 22.8 mL of MeOH and 4.3 mL of water (MeOH/water = 3/1 v/v). Copolymerization was carried out at 70 °C for 24 h and then stopped by quenching the flask to liquid nitrogen. The copolymers were purified by dialysis through a 12–14 kDa MWCO cellulose membrane against water, and the copolymer was obtained by freeze-drying, giving the white powder (practical yield 60 %).

### 2.3. RAFT copolymerization of MPC and PEOMEMA in NMR tubes

For the kinetics study, copolymerization of MPC and PEOMEMA in a mixture of deuterated solvents MeOD and $D_2O$ (3/1 v/v) was carried out in NMR tubes. 0.1329 g of MPC (0.45 mmol), 0.855 mL of PEOMEMA solution (0.45 mmol), 2.6 mg of CTA (0.9 μmol) and 0.8 mg of ACVA (0.3 μmol) were placed in 10 mL round-bottom flask and dissolved in 5.2 mL of MeOD and 0.97 mL of $D_2O$. The solution was distributed to NMR tubes equally, and each tube was purged with Ar gas for 30 min, then sealed, and the tubes were put into the oven at 70 °C. After a certain time, the tubes were taken and quenched to liquid nitrogen. The tube with the solution was analyzed by NMR, then 0.1 mL of the solution was taken and diluted to 1.5 mL by MeOH/water solution (3/1 v/v), and the diluted solution was used for *SEC* analysis.

To determine the reactivity ratios of MPC and PEOMEMA, copolymerization of MPC and PEOMEMA was carried out in NMR tubes. Components of the reaction mixture were dissolved previously in deuterated solvents and dosed to NMR tubes as 10 % MPC solution in MeOD, 10 % CTA solution in MeOD and 1 % ACVA solution in MeOD. For copolymerization of MPC and PEOMEMA at molar ratio 40:60, the required ingredients were dosed to NMR tube as follows: 0.1193 mL of MPC solution (32 μmol), 0.0912 mL of aqueous (50 %) PEOMEMA solution (48 μmol), 2.9 μL of GPA solution (0.80 μmol), 9.4 μL of ACVA solution (0.266 μmol), 0.402 mL of MeOD and 0.092 mL of $D_2O$. The tubes were purged with Ar gas for 30 min, sealed and analyzed by NMR. Then the copolymerization was carried out at 70 °C for 2 h, stopped by quenching tubes to liquid nitrogen, and the tubes with the solutions were analyzed by NMR again.

### 2.4. Analysis and characterization

$^1$H and $^{13}$C NMR spectra were recorded on a *Bruker 400 Ascend*™ nuclear magnetic resonance spectrometer (400 MHz) in $D_2O$ at 22 °C. Overall (sum) monomer conversion (mol.%) during RAFT copolymerization was calculated by the eq. (1):

$$q^t = \frac{\left(\int_{5.71}^{0} + \int_{6.16}^{0}\right) - \left(\int_{5.71}^{t} + \int_{6.16}^{t}\right)}{\int_{5.71}^{0} + \int_{6.16}^{0}} \bullet 100 \tag{1}$$

where $\int_{6.16}^{t}$ and $\int_{5.71}^{t}$ are integrals of the vinyl protons shifts in $^1$H NMR spectra of the monomers at 6.16 ppm and 5.71 ppm, respectively, after a time *t* of copolymerization, and $\int_{6.16}^{0}$ and $\int_{5.71}^{0}$ are the same shifts in $^1$H NMR spectra before copolymerization.

Conversion of MPC (mol.%) was calculated by eq. (2):

$$q^t_{MPC} = \frac{\int_{3.25}^{0} - \int_{3.25}}{\int_{3.25}^{0}} \cdot 100 \tag{2}$$

where $\int_{3.25}^{0}$ and $\int_{3.25}^{t}$ are integrals of the protons shifts in -N(C$\underline{H}_3$)$_3$) group of the MPC monomer in $^1$H NMR spectra before copolymerization and after a time *t* of copolymerization, respectively.

Conversion of PEOMEMA (mol.%) was calculated by Eq. (3):

$$q^t_{PEOMEMA} = \frac{q^t - f^0_{MPC} \bullet q^t_{MPC}}{f^0_{PEOMEMA}} \bullet 100 \tag{3}$$

where $q^t$ is the sum monomer conversion (Eq. (1)), $q^t_{MPC}$ is the MPC conversion (Eq. (2)), $f^0_{MPC}$ and $f^0_{PEOMEMA}$ are molar fractions of MPC and PEOMEMA in the initial monomer mixture, respectively.

The average composition of the copolymer after a specific time *t* of copolymerization was calculated by Eq. (4):

$$F^t_{MPC} = \frac{q^t_{MPC}}{q^t_{MPC} + q^t_{PEOMEMA}}; F^t_{PEOMEMA} = \frac{q^t_{PEOMEMA}}{q^t_{MPC} + q^t_{PEOMEMA}} \tag{4}$$

The composition of the monomer mixture before copolymerization

(Eq. (5)) and at a specific time $t$ of copolymerization was calculated by Eq. (6):

$$f^0_{MPC} = \frac{\frac{1}{3}\int^0_{3.25}}{\int^0_{3.39} + \frac{1}{3}\int^0_{3.25}}; f^0_{PEO} = \frac{\int^0_{3.39}}{\int^0_{3.39} + \frac{1}{3}\int^0_{3.25}}; f^0_{MPC} = \frac{\int^0_{4.14}}{\int^0_{3.79} + \int^0_{4.14}}; f^0_{PEO} = \frac{\int^0_{3.79}}{\int^0_{3.79} + \int^0_{4.14}} \quad (5)$$

where $\int^0_{3.39}$, $\int^0_{3.79}$ and $\int^0_{4.14}$ are integrals of the protons shifts in terminal –$OCH_3$ group in PEOMEMA, –COO-$CH_2$-C$\underline{H}_2$-O- group in PEOMEMA, and -O-$CH_2$-C$\underline{H}_2$—O—P- group in MPC, respectively;

$$f^t_{MPC} = f^0_{MPC} - f^0_{MPC} \bullet q^t_{MPC}; f^t_{PEOMEMA} = f^0_{PEOMEMA} - f^0_{PEOMEMA} \bullet q^t_{PEOMEMA} \quad (6)$$

Instantaneous copolymer composition was calculated by Eq. (7):

$$F^{MPC}_{inst.} = \frac{\Delta f^t_{MPC}}{\Delta f^t_{MPC} + \Delta f^t_{PEOMEMA}} \quad (7)$$

The reactivity ratios of MPC and PEOMEMA in RAFT copolymerization were calculated by the methods of Fineman and Ross [23], Kelen and Tüdos [24], and Meyer and Lowry [25]. Theoretical (based on the terminal model of free-radical copolymerization) dependencies of instantaneous monomer feed composition and instantaneous copolymer composition on conversion were calculated by equations presented elsewhere [26].

The number average molecular weight ($M_n$), dispersity index ($Đ = M_w/M_n$), monomer conversion, copolymerization rate and average copolymer composition were determined by size exclusion chromatography (*SEC*). *SEC* measurements were carried out in MeOH/$H_2O$ (3/1 v/v) as an eluent at 30 °C using a constant flow rate of 0.5 mL/min; column Viscotek T6000 M General Mixed, 300 × 8.0 mm. Viscotek TDAmax (Malvern) system equipped with a triple detection array (TDA305) consisting of a refractive index detector (RI), light scattering detector (LS) simultaneously measuring the scattered light (laser 3 mW, 670 nm) at two angles – right-angle (90°) and low-angle (7°), and four-capillary bridge viscosity detector (DP) and a Viscotek UV detector 2500 (UV) was used.

The hydrophilicity of the copolymers was studied by measuring the water contact angle (WCA) on the copolymer layer using *Theta Lite Optical Tensiometer* from Biolin Scientific (Finland). WCA was measured in 3 different spots on the copolymer-coated glass surface. Before the experiment, the glass substrates were cleaned according to the method presented before [27]: the glass slides were rinsed by deionized water, then incubated for 1 h in conc. sulfuric acid, washed with deionized water and with methanol, and dried under $N_2$ flow. The glass slides were immersed into methanol copolymer solutions (0.01 wt%) for 24 h, then pulled out and dried without rinsing. A drop of deionized water (10 μL) was applied onto the copolymer layer, and after 30 s the picture of the drop was captured and analyzed.

DLS measurements of the copolymer solutions were carried out using *Zetasizer Nano ZS* (Malvern Instruments) equipped with a 4 mW HeNe laser at a wavelength of 633 nm. Measurements of the intensity of the scattered light were performed at an angle of 173°. The size distribution data were analyzed using Zetasizer software (v.7.12). The copolymer concentration in ethanol/water solutions was kept at 10 mg/mL.

## 3. Results and discussion

### 3.1. Kinetics of the RAFT copolymerization of MPC and PEOMEMA

Synthetic route to the copolymers p(MPC-*co*-PEOMEMA) via RAFT copolymerization of MPC and PEOMEMA is presented in Fig. 1a. It is well established [22,28,29] that RAFT copolymerization could give either random or gradient brush copolymer. The architecture of the brush copolymers depends on the reactivity ratios of the monomers, which could be evaluated by studying the kinetics of the copolymerization. The arrangement of PEO side chains along the backbone of the copolymers p(MPC-*co*-PEOMEMA) considerably affects the accessibility of phosphoryl choline groups and hydration of the copolymers. Kinetics of the RAFT copolymerization of MPC and PEOMEMA was studied by $^1$H

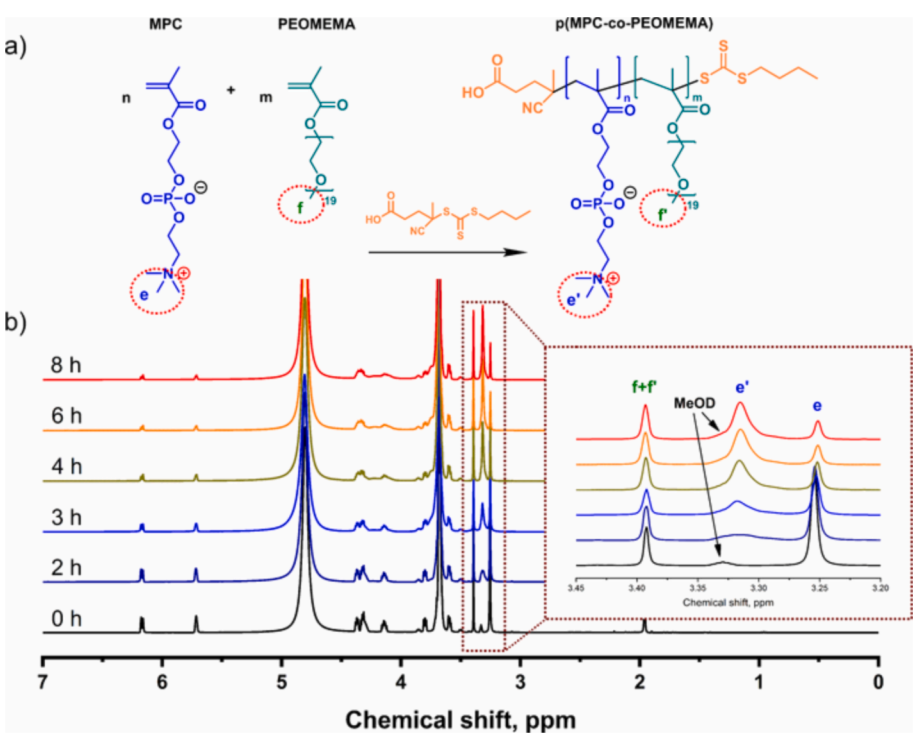


**Fig. 1.** Scheme of the RAFT copolymerization of MPC and PEOMEMA with attribution protons to the chemical shifts in $^1$H NMR spectra (a); $^1$H NMR spectra of the reaction mixture taken during RAFT copolymerization of MPC and PEOMEMA in MeOD:$D_2O$ (3:1 v/v); $[MPC]_0$:$[PEOMEMA]_0$ = 53:47 mol.% (b); insert right – an expanded region of the spectra at 3.15–3.45 ppm.

NMR spectroscopy (Fig. 1b) and size exclusion chromatography (*SEC*) using a methodology similar to that published in our previous papers [22]. Copolymerization of MPC and PEOMEMA in the mixed solvent MeOH:water = 3:1 v/v (or in MeOD:$D_2O$ = 3:1 v/v, if the copolymerization was carried out in NMR tubes) was carried out for a certain time, and aliquots of the reaction mixture were taken during the first 8 h for kinetics study. The eluent for *SEC* analysis was identical to the solvent for copolymerization, i.e. MeOH:water = 3:1 v/v.

First of all, the kinetics of the RAFT copolymerization of MPC and PEOMEMA was studied by analyzing the reaction mixtures by $^1$H NMR spectroscopy. Changes in chemical shifts and their intensity in $^1$H NMR spectra during copolymerization of the equimolar initial monomer mixture are presented in Fig. 1b. In $^1$H NMR spectra, the shifts at 5.71 ppm, 6.16 ppm, and 6.17 ppm are attributed to the vinyl protons of the monomers. Unfortunately, the chemical shifts of the vinyl protons of both monomers are almost at the same positions, which does not allow using them for the calculation of the consumption of the individual monomers. Nevertheless, a decrease in their intensity during copolymerization could be used to evaluate the overall monomer conversion. The shift at 3.39 ppm is attributed to the protons of the terminal –$OCH_3$ group present in both the PEOMEMA monomer and PEOMEMA units in the copolymers. On the contrary, the shift of the protons attributed to the group –$N(CH_3)_3$ changed the position from 3.25 ppm in MPC monomer to 3.31 ppm in MPC monomeric unit in the copolymer during copolymerization. The chemical shift at 3.31 ppm is unsuitable for analytical purposes since it overlaps with the chemical shift of the deuterated methanol. Fortunately, a decrease in intensity of the shift at 3.25 ppm attributed to MPC monomer in combination with a decrease in intensity of the vinyl protons attributed to both monomers during copolymerization allowed to calculate the consumption of the individual monomers and, at the same time, the average composition of the copolymer.

An analysis of the initial reaction mixtures with various monomer ratios by $^1$H NMR spectroscopy revealed that the actual composition of the monomer mixtures was slightly different. For example, for the targeted equimolar monomer mixture, the actual monomer mixture was [MPC]:[PEOMEMA] = 53.6:46.4 mol.% (according to the integrated shifts at 3.25 ppm and 3.39 ppm attributed to the protons of –$N(CH_3)_3$ group in MPC and -O-$CH_3$ group in PEOMEMA, respectively), and 52.5:47.5 mol.% (according to the integrated shifts at 4.14 ppm and 3.79 ppm attributed to the protons of -O-$CH_2$-C$\underline{H}_2$—O—P- group in MPC and –COO-$CH_2$-C$\underline{H}_2$-O- group in PEOMEMA, respectively). Average composition of that monomer mixture was considered as [MPC]:[PEOMEMA] = 53:47 mol.%; a certain deviation from the target composition ([MPC]:[PEOMEMA] = 50:50 mol.%) was predetermined by inaccuracy in preparing small amount of the reaction mixture in NMR tube.

Copolymerization of MPC and PEOMEMA at different monomer ratios was carried out in deuterated solvents in NMR tubes to relatively low conversions (q = 3–21 %); data of these copolymerizations, including overall monomer conversions and copolymer compositions, are presented in ESI. Compositions of the monomer mixtures as well as copolymer compositions at early stages of copolymerization were calculated from $^1$H NMR spectra of the reaction mixtures. The dependence of the copolymer composition $F_1$ on the initial comonomer feed composition $f_1$ for the RAFT copolymerization of MPC ($M_1$) and PEOMEMA is presented in Fig. 2. Experimental points displayed in Fig. 2 evidently show that the copolymers are consistently enriched in MPC units. Reactivity ratios $r_1$ and $r_2$ for the comonomer pair MPC ($M_1$) and PEOMEMA ($M_2$) were calculated by several methods and are presented in Table 1.

Calculation according to the method of Meyer and Lowry was realized by iterative recalculation of the copolymer composition $F_1$ using initial $f_0$ and shifted $f_1$ monomer feed compositions (shifted value of $f_1$ depends on monomer conversion according to the equation of Meyer and Lowry). Iterations were set until the values of $r_1$ and $r_2$ were changing insignificantly. Illustration of determination of the reactivity ratios of the monomers MPC ($M_1$) and PEOMEMA by the method of Meyer and Lowry is presented in ESI. One should note that the monomer reactivity ratios only slightly depend on the calculation method. This is not unexpected, since overall monomer conversion was rather low (0.03 to 0.21). Nevertheless, the most accurate values of $r_1$ and $r_2$ were determined by the method of Meyer and Lowry, where the changes of the monomer feed composition with conversion were taken into account. The monomer reactivity ratios $r_1$ = 2.48 and $r_2$ = 0.40 determined by the method of Meyer and Lowry were further used for calculation of the theoretical dependences $F_1$ on q. Much higher reactivity ratio of MPC compared to that of PEOMEMA implies that MPC is consumed faster during copolymerization, which affects the comonomer feed composition and, consequently, the microstructure of the copolymers.

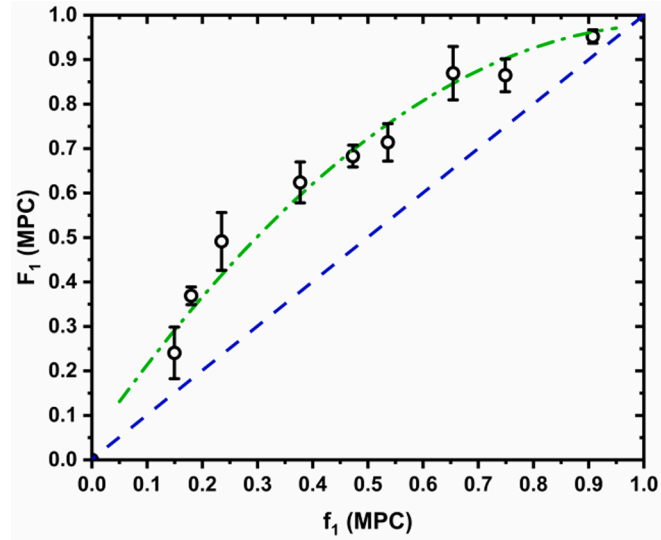


**Fig. 2.** Theoretical (dashed dotted curve) and experimental (points with error bars) dependences of the copolymer composition $F_1$ on the initial comonomer feed composition $f_1$ for the RAFT copolymerization of MPC ($M_1$) and PEOMEMA. Theoretical dependence was calculated using reactivity ratios of the monomers $r_1$ = 2.48, $r_2$ = 0.40, determined by the method of Mayer-Lowry.

**Table 1**
Reactivity ratios of MPC ($M_1$) and PEOMEMA ($M_2$) calculated by various methods.

| Method | $r_1$ | $r_2$ |
|---|---|---|
| Fineman-Ross [23] | 2.10 ± 0.28 | 0.32 ± 0.08 |
| Kelen-Tudos [24] | 2.28 ± 0.22 | 0.27 ± 0.09 |
| Meyer-Lowry [25] | 2.48 | 0.40 |

It was shown in our previous publication [22], that kinetics of copolymerization can be studied using *SEC* data if one of the monomers possesses relatively high molecular weight. *SEC* eluograms of the reaction mixtures taken during RAFT copolymerization of MPC and PEOMEMA are presented in Fig. 3. The RI response signals attributed to the monomer MPC, macromonomer PEOMEMA, and the copolymer are slightly overlapped. The evolution of the *sEC*-RI eluograms during copolymerization evidently show a decrease in intensity of the monomer and the macromonomer signals and, at the same time, an increase and a shift of the copolymer signal to lower retention volume (higher molecular weight). The signals attributed to the monomer, the macromonomer, and the copolymer were deconvoluted by the method described before [22]; deconvoluted *SEC* eluograms are presented in ESI. Deconvoluted signals (their integrals) were used to calculate the overall monomer conversion, copolymerization rate, and copolymer composition [22].

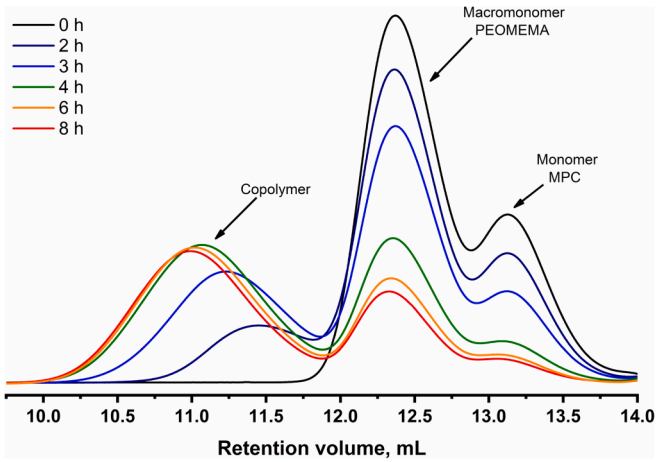


**Fig. 3.** *sEC*-RI eluograms of the reaction mixture taken during RAFT copolymerization of MPC and PEOMEMA. Eluent MeOD:$D_2O$ (3:1, v/v) containing 0.1 M $NaNO_3$; $[MPC]_0$:$[PEOMEMA]_0$ = 53:47 mol.%.

An analysis of the same initial reaction mixture with equimolar monomer ratio by $^1$H NMR spectroscopy and by *SEC* gave unexpected result: according to $^1$H NMR spectroscopy, MPC content in the initial mixture was 53 mol.% while according to *SEC* – 69 mol.%. Such a significant difference in composition of the initial monomer mixture determined by two independent techniques cannot be explained by the inaccuracy of the methods. Trying to explain the phenomenon, we made a presumption that the *sEC*-RI signal of the monomer MPC depends on the concentration of the monomer plus the concentration of the bound water molecules. It is well established [30–32] that MPC is highly hydrated, thus the phosphorylcholine side group is surrounded by water molecules and, carry" them as an integral part of the monomer. The molecular weight of the monomer MPC is 295. We tried to increase the molecular weight of the hydrated MPC by increasing the number of bound water molecules and continuing to calculate the copolymer composition. The composition of the initial monomer mixture in *SEC* analysis was the same as determined by $^1$H NMR spectroscopy ([MPC]:[PEOMEMA] = 53:47 mol.%) if the molecular weight of MPC was set to 565. These calculations led to the simple conclusion that every molecule of MPC carries about 15 molecules of water as an integral part of the monomer, at least, in methanol:water (3:1 v/v) solutions.

To verify the correctness of the presumption about the "increased" molecular weight of MPC shown by *sEC*-RI curves, we calculated average copolymer composition at various monomer conversions from $^1$H NMR spectra of the reaction mixtures, and also from *SEC* data of the same mixtures using two molecular weights of MPC – 295 and 565. Changes in the average copolymer composition (amount of MPC units) with the sum conversion of the monomers are presented in Fig. 4 (a). Obviously, the curves 2 and 3 illustrating the dependences based on $^1$H NMR spectra and *SEC* data with MPC of molecular weight 565 are very close. Contrarily, the curve based on *SEC* data with MPC of molecular weight 295 is shifted towards a very high content of MPC units in the copolymers, which is even impossible at high conversions of the monomers. Thus, our experiments prove that in the solutions of methanol and water (3:1 v/v) one molecule of the monomer MPC carries about 15 molecules of water.

Water molecules have been shown to form a cage structure around zwitterionic polymers, similar to the molecular organization of ice [33]. Water molecules are attracted to the charged functional groups of the zwitterion through ionic solvation, which forms an energy barrier which is greater than that associated with a poly(ethylene glycol) hydration shell, which is formed through hydrogen bonding. Although PEO is abundantly hydrated, such hydrates are less organized allowing the water exchange. As it was shown in our previous study [34], apparent molecular weight of PEOMEMA was not increasing due to hydration. An increase in apparent molecular weight of MPC in methanol–water solutions we attribute to formation of clathrate cage structure of surrounding water molecules, in which water molecules are retained like in ice.

Monomer consumption curves during RAFT copolymerization of MPC and PEOMEMA in the mixed solvent of methanol and water are presented in ESI. Conversion of the monomers depends on the monomer ratio being higher at excess of MPC. Overall conversion of the monomers after 8 h of copolymerization is moderately high, about 70–80 mol.%. The pseudo-first order kinetic plot for the RAFT copolymerization of MPC and PEOMEMA, plotted through experimental points obtained by $^1$H NMR spectroscopy and by *SEC*, is linear (refer to ESI). The same linear dependence for the points obtained by two independent methods

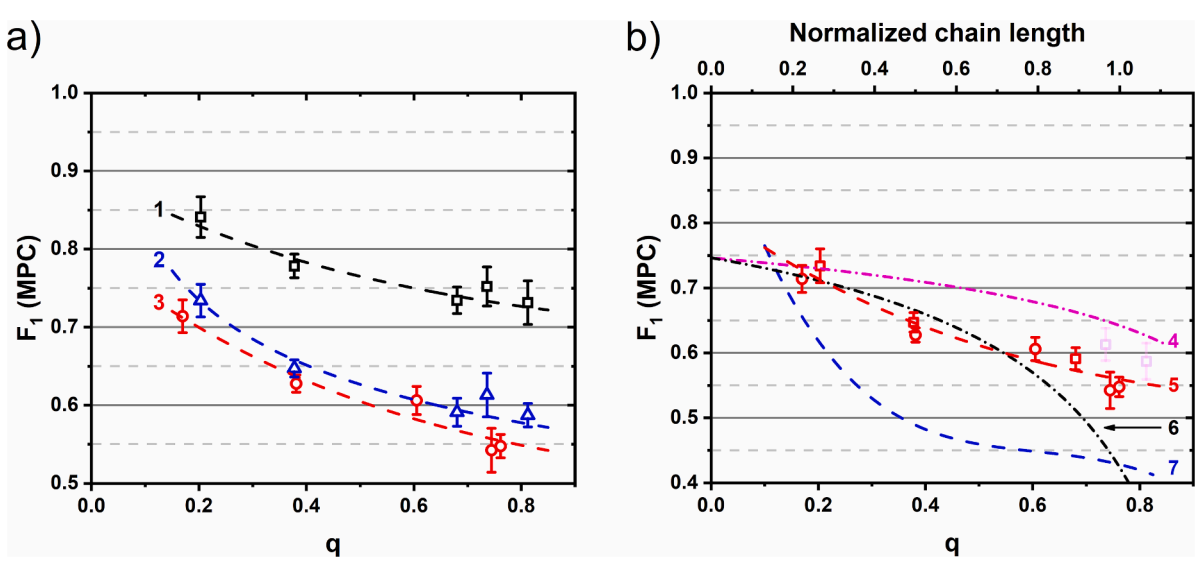


**Fig. 4.** (a) Changes in average copolymer composition during RAFT copolymerization of MPC and PEOMEMA calculated from $^1$H NMR spectra (3) and *sEC*-RI data (1, 2) using MPC with molecular weight 295 (1) and 565 (2); (b) changes in average copolymer composition during RAFT copolymerization of MPC and PEOMEMA calculated from experimental data (5) of *sEC*-RI (○) and $^1$H NMR spectra (□) and theoretically using reactivity ratios of the monomers $r_1$ = 2.48, $r_2$ = 0.40 (4); (b) evolution of instantaneous copolymer composition formed at a certain overall monomer conversion as a function of the apparent normalized chain length calculated by derivatization of the curve 2 (7) and theoretically using reactivity ratios of the monomers $r_1$ = 2.48, $r_2$ = 0.40 (6); $[MPC]_0$:$[PEOMEMA]_0$ = 53:47 mol.%.

demonstrates that *SEC* is a suitable method to study the kinetics of RAFT copolymerization. The copolymerization rate constant calculated from the slope of the plot is $8.46 \times 10^{-5}\ s^{-1}$. Thus, the rate of the RAFT copolymerization of MPC and PEOMEMA is moderately high.

An average (cumulative) copolymer composition determined by *SEC* and by NMR spectroscopy is very close, and its changes with conversion are represented by the same curve (Fig. 4b, curve 5). Short chains of the copolymers formed at the initial stages of the copolymerization (at relatively low monomer conversions) are rich in MPC units. In contrast, long polymer chains (formed at high monomer conversions) contain nearly the same amount of MPC and PEOMEMA units. The course of the curve presenting changes in average copolymer composition, calculated theoretically according to the reactivity ratios of the monomers (Fig. 4b, curve 4), is different from the curve 5: theoretically calculated copolymer composition is changing much less; moreover, it is characterized by upward curvature, in contrast to the experimental curve. A comparison of these two curves suggests that experimental data are poorly described by the theoretical model based on the reactivity ratios of the monomers.

An average composition of the copolymer synthesized by RAFT copolymerization does not reflect the real composition of a chain fragment formed at a certain conversion point (at a certain chain length). Changes in copolymer composition along polymer chains are presented by the evolution of instantaneous copolymer composition during chain growth. Instantaneous copolymer composition as a function of the apparent normalized chain length calculated by derivatization of the curve presenting changes in average copolymer composition during copolymerization and calculated theoretically using experimentally determined reactivity ratios of the monomers are presented in Fig. 4b, curves 7 and 6, respectively. Obviously, changes in instantaneous copolymer composition along polymer chains are much larger compared to changes in average copolymer composition during copolymerization. For instance, for the copolymers synthesized by RAFT copolymerization of nearly equimolar monomer mixture to overall monomer conversion of about 80 mol.%, instantaneous copolymer composition changes along polymer chains from 75 mol.% of MPC units at the beginning to about 40 mol.% of MPC units at the end of the chains (Fig. 4b, curves 6 and 7). Thus, the copolymers of MPC and PEOMEMA synthesized by RAFT copolymerization are characterized by clearly expressed gradient structure.

It is worth noting that the course of the two curves 6 and 7 (Fig. 4b) presenting the evolution of instantaneous copolymer composition is very different, i.e., the theoretically calculated curve is poorly coinciding with the curve obtained by derivatization of the experimental data. It means that the microstructure (changes in copolymer composition along polymer chains) of the copolymers p(MPC-*co*-PEOMEMA) synthesized by RAFT is not described (or poorly described) by the terminal model of free-radical copolymerization based on reactivity ratios of the monomers.

Although the monomer reactivity ratios in reversible deactivation radical copolymerizations are generally the same as in free-radical copolymerizations, there is a difference in the distribution of copolymer compositions for different polymer molecules [35,36]. It was shown by Matyjaszewski [37] that the differences arise not from differences in monomer reactivity ratios, but from differences in the consumption of the two monomers. The effect is observed in the copolymerization systems where there is an intermittent activation/deactivation process between active and dormant species. Deviations are possible when two propagating species have different activation and/or deactivation rate constants. Under these conditions, one of the monomers is consumed faster compared to corresponding free-radical system, and the comonomer feed and, therefore, the copolymer composition as a function of conversion changes differently than in free-radical polymerization. The effect is generally larger when homopropagation is faster than cross-propagation ($r_1$, $r_2 > 1$) and at lower monomer conversion.

Likely, the above-described conditions are realized in the RAFT copolymerization of MPC and PEOMEMA (equilibrium between active and dormant species, propagating species containing a terminal unit of MPC or PEOMEMA have different activation and/or deactivation rate constants), and the monomer MPC is consumed considerably faster. This leads to a swift change in instantaneous copolymer composition during RAFT copolymerization of MPC and PEOMEMA. Consequently, even copolymers of MPC and PEOMEMA synthesized by RAFT copolymerization to the medium conversion of the monomers possess clearly expressed gradient microstructure.

The recent review compares and contrasts the properties of gradient copolymers with properties of the analogous block and statistical copolymers [38]. The most significant differences were found in interfacial, thermal and micellization properties of the copolymers. Gradient copolymers produce broad interfaces with no distinct boundaries and show unusually broad glass transitions. The gradient copolymer localizing at the interface lowers the interfacial tension and suppresses the phase separation process. Gradient copolymers have a higher solubility than the corresponding diblock copolymers, leading to significantly higher values of the CMC. One of the most commonly reported applications is as compatibilizers for immiscible polymer blends. Other applications reported for gradient copolymers include as stabilizers of emulsions or dispersions, as thermoplastic elastomers, and multi-shape memory materials [38].

Fig. 5 demonstrates changes in molecular weight and dispersity of the copolymers during RAFT copolymerization of MPC and PEOMEMA. MWD curves of the copolymers are presented ESI. They are unimodal and symmetrical without shoulders and tailing; their shift toward higher molecular weight with conversion illustrates an increase in molecular weight of the copolymers. Experimentally determined molecular weights correspond well with those calculated theoretically. Theoretically calculated dependence of molecular weight on monomer conversion is presented by a bent line, which is a product of uneven incorporation of the monomers into the growing polymeric chains during copolymerization (gradient distribution of the monomeric units across the polymeric chains). Dispersity of the copolymers Đ is not low but moderate, which is predetermined mainly by steric effects of the macromonomer PEOMEMA. Dispersity of the copolymers containing less PEOMEMA units (lower density of PEO side chains) is lower (see ESI), which is in correlation with previously published data [39]. At a certain point of the chain growth (q about 60 %), the dispersity Đ of the copolymers becomes almost independent of the monomer conversion (chain length). Thus, RAFT copolymerization of MPC and PEOMEMA in

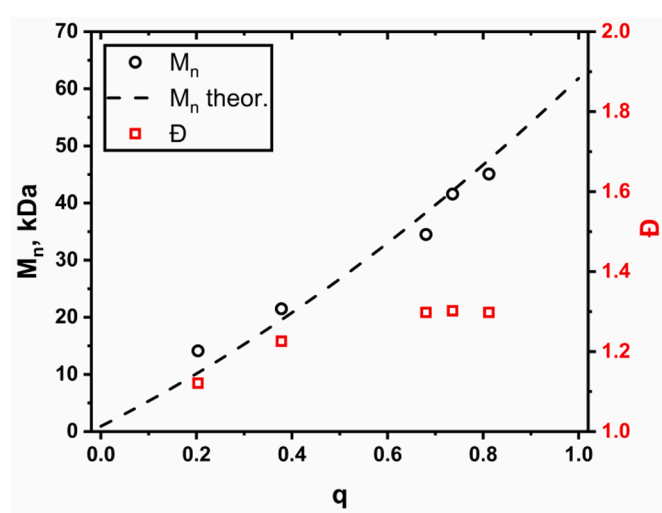


**Fig. 5.** Evolution of molecular weight (○) and dispersity (□) of the copolymers during RAFT copolymerization of MPC and PEOMEMA; $[MPC]_0$:$[PEOMEMA]_0$ = 53:47 mol.%. Dashed line – theoretical (calculated) dependence of the molecular weight of the copolymers on conversion.

the presence of trithiocarbonate-type CTA proceeded in reasonably controlled manner giving well-defined gradient copolymers with predicted macromolecular parameters, i.e., molecular weight close to theoretically calculated and moderately low dispersity (Đ below 1.3).

### 3.2. Some properties of the gradient copolymers of MPC and PEOMEMA

A series of the copolymers p(MPC-*co*-PEOMEMA) differing in composition and molecular weight were synthesized (Table 2). Copolymerization for 24 h yielded copolymers with degree of polymerization (DP) 60–70 and dispersity index Đ 1.2–1.3. The highest dispersity (Đ 1.30) was characteristic for the brush copolymer containing high content of PEOMEMA units. The composition of the copolymers determined by $^{1}$H NMR spectroscopy was different from the composition of the corresponding monomer feed confirming much higher reactivity of MPC determined studying kinetics of copolymerization. In order to ascertain the effect of PEOMEMA units on characteristic properties of MPC copolymers, hydrophilicity and cononsolvency of the copolymers differing in composition were evaluated.

It is well known that pMPC is superhydrophilic material with a water contact angle (WCA) of less than 20° [40–42]. pPEOMEMA is also hydrophilic and like pMPC is capable of holding water molecules. The hydrophilicity of the copolymers p(MPC-*co*-PEOMEMA) containing a small and large amounts of MPC units was evaluated by measuring WCA on the adsorbed copolymer layers formed on glass slides when these slides were kept immersed in dilute copolymer solutions, then pulled out and dried. The WCA of the bare glass surface is ≈30° [43]. Coating of the glass surface with the copolymers p(MPC-*co*-PEOMEMA) increases the hydrophilicity shifting WCA of the surface from 35° to as low as 7° (Fig. 6). It is worth to mention that the WCA of the copolymers does not depend on copolymer composition and is very low, which means that the copolymers p(MPC-*co*-PEOMEMA) are superhydrophilic. A comparison of WCA values for pMPC and the copolymers p(MPC-*co*-PEOMEMA) shows the dominant role of MPC units in determining the hydrophilicity of the copolymers.

It is well known that pMPC exhibits so called cononsolvency, i.e., it is soluble both in water and ethanol, but is insoluble in the mixtures of the above solvents containing 60–92 vol% of ethanol [30,44]. Cononsolvency of pMPC is explained by competitive hydrogen bonding of the solvent and cosolvent with the polymer: in this hydrogen bond competition pMPC side groups prefer water rather than ethanol since water molecules form an ice-like clathrate structure around the MPC side group facilitating the formation of the inner salt [30].

Cononsolvency of the copolymers p(MPC-*co*-PEOMEMA) was studied by DLS. The solubility of the copolymers was evaluated by measuring the hydrodynamic diameter ($D_h$) of the particles formed in water, ethanol, and their mixtures. DLS measurements were carried out at various temperatures, which made it possible to evaluate the effect of two factors (solvent and temperature) on solubility, macromolecular size, and aggregation of the copolymers.

Particle size distribution (PSD) curves of the copolymer p(MPC-*co*-PEOMEMA) containing 77 mol.% of MPC units in water and ethanol are presented in Fig. 7. Analogous curves for the copolymer containing 32 mol.% of MPC units are presented in ESI. Obviously, water and ethanol are good solvents for the copolymers p(MPC-*co*-PEOMEMA),

**Table 2**
Characteristics of the copolymers p(MPC-*co*-PEOMEMA). $[M]_0$:$[CTA]_0$:$[ACVA]_0$ = 240:3:1, τ = 24 h.

| Sample | $[MPC]_0$/ $[PEOMEMA]_0$ | Practical yield, % | Copolymer composition, MPC mol.% | $M_n$ | DP | Đ |
|---|---|---|---|---|---|---|
| 1 | 25:75 | 51 | 31.9 | 39,400 | 61 | 1.30 |
| 2 | 50:50 | 64 | 59.7 | 34,100 | 65 | 1.25 |
| 3 | 75:25 | 70 | 77.4 | 33,700 | 73 | 1.19 |

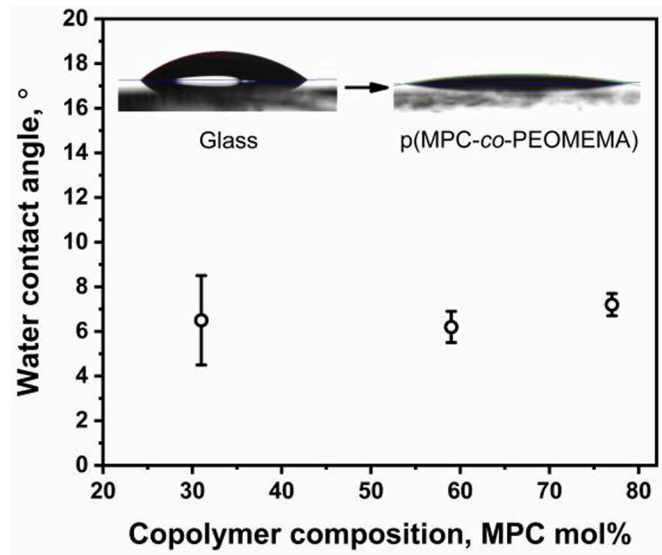


**Fig. 6.** Water contact angle of the copolymers p(MPC-*co*-PEOMEMA). Insert – visualization of the determination of WCA.

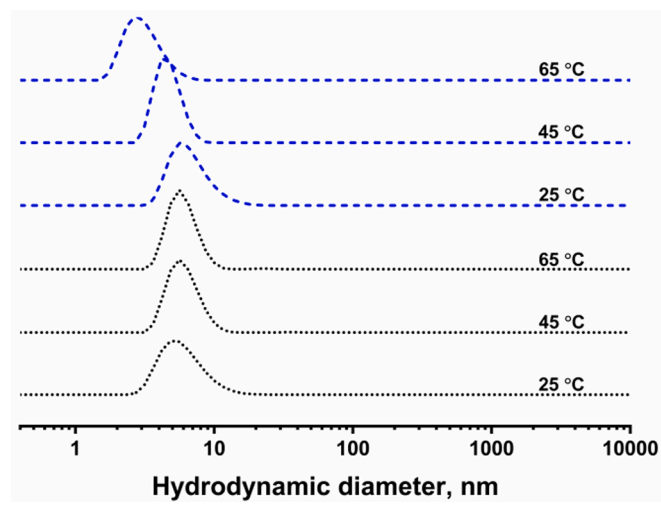


**Fig. 7.** PSD curves (by volume) of the copolymer p(MPC-*co*-PEOMEMA) containing 77 mol% of MPC units in water (black) and ethanol (blue) at various temperatures.

irrespective of their composition, since the copolymers are at the molecular level with a hydrodynamic diameter of the particles less than 10 nm. Elevation of temperature did not affect the particle size of the copolymers dissolved in pure water, e.g., the particle size of the copolymer p(MPC-*co*-PEOMEMA) containing 77 mol.% of MPC units remained almost the same 6.4 ± 2.1 nm at 25 °C and 5.7 ± 1.3 nm at 65 °C (Fig. 7). In contrast, the diameter of the particles of the same copolymer dissolved in ethanol was reduced almost twice by increasing the temperature from 25 °C (d = 4.3 ± 1.2 nm) to 65 °C (d = 2.3 ± 0.59 nm). Compactization of the particles at higher temperature indicates that solubilizing power of ethanol is becoming worse under these conditions; nevertheless, the copolymers are still soluble in ethanol at 65 °C. Solvation of the copolymers p(MPC-*co*-PEOMEMA) by water is very good and does not depend on temperature.

Solubility of the copolymers in the mixed solvent of water and ethanol with an ethanol content of 20 vol% and 40 vol% was also good, with the particle size diameter of less than 10 nm (Fig. 8). Addition of ethanol reduced the hydrodynamic diameter of the particles to 2–4 nm, which evidently shows worse solvating conditions in ethanol-containing

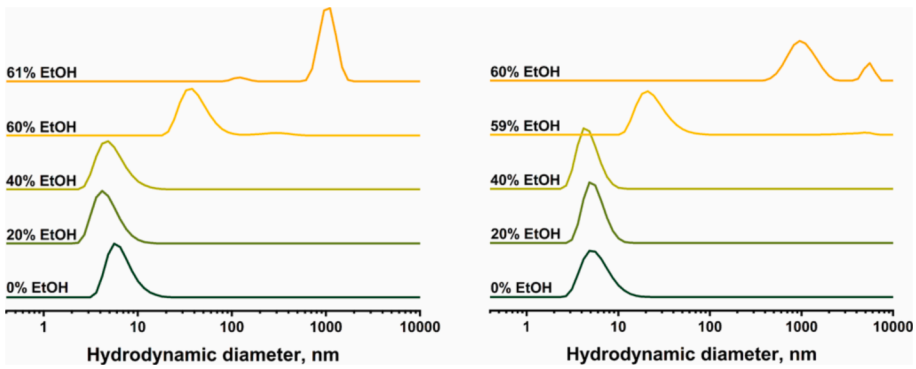


**Fig. 8.** PSD curves (by volume) of the copolymer p(MPC-*co*-PEOMEMA) containing 32 mol.% (left) and 77 mol.% (right) of MPC units in water and water–ethanol mixtures.

solutions. This is in accord with the explanations of the cononsolvency of pMPC based on relatively worse solvating of the polymer by ethanol [30]. In the mixed solutions containing about 60 vol% of ethanol, aggregation of the copolymers p(MPC-*co*-PEOMEMA) occur, which is reflected in a sharp increase in particle size (Fig. 8). In this case, there is a slight difference between the copolymers, containing large (77 mol.%) and small (32 mol.%) amount of MPC units: the copolymers with a large amount of MPC units begin to aggregate when ethanol content in the mixed solvent is 59 vol%, while the copolymers with small amount of MPC units – when ethanol content is 60 vol%. At such compositions of the mixed solvent, small but stable (micellar?) aggregates are formed (d = 20–40 nm), which are converted to large aggregates (d > 800 nm) at higher content of ethanol in the mixed solvent.

## 4. Conclusions

Kinetics of the RAFT copolymerization of the zwitterionic monomer 2-methacryloyloxyethyl phosphorylcholine (MPC, $M_1$) with the macromonomer PEO methacrylate with medium-length PEO chain (PEOMEMA, $M_2$) was studied by two methods $^1$H NMR spectroscopy and size exclusion chromatography (*SEC*) which allowed evaluating the distribution of copolymer compositions. The dependences of average (cumulative) copolymer composition on the conversion of the monomers determined by the two methods were coincident in between but differed significantly from that calculated by the terminal model of free-radical copolymerization based on reactivity ratios of the monomers determined experimentally $r_1 = 2.48$ and $r_2 = 0.40$. Even larger differences were between compositions of the copolymer fragments, formed at a certain overall monomer conversion (at a certain point of the copolymer chain) calculated by derivatization of the experimental curve and by the model. The data are in accord with theoretical predictions, which are scarcely proved experimentally, that the distribution of copolymer compositions for reversible deactivation radical copolymerization and free-radical copolymerization may be different and depends on activation and/or deactivation rate constants of the competitive propagating species.

The copolymers p(MPC-*co*-PEOMEMA) possessed strongly expressed gradient microstructure, even those synthesized up to medium conversion of the monomers. Despite gradient microstructure, dispersity of the copolymers was moderately low (Đ < 1.3), and molecular weight was close to that predicted theoretically. The copolymers p(MPC-*co*-PEOMEMA), irrespective of composition, showed evident superhydrophilicity with a water contact angle about 7°. The copolymers exhibited cononsolvency in water–ethanol mixed solutions, very similar to that for pMPC.

In mixed solutions of methanol and water, every molecule of MPC carries about 15 molecules of water as an integral part of the monomer; this should be taken into account employing *SEC* technique to study copolymerization kinetics.

## CRediT authorship contribution statement

**Marijus Jurkūnas:** Investigation, Data curation, Visualization, Writing – original draft. **Vaidas Klimkevičius:** Methodology, Validation, Data curation, Writing – original draft. **Agnė Uscilaitė:** Investigation, Data curation. **Ričardas Makuška:** Supervision, Conceptualization, Validation, Writing – review & editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2022.111764.

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